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Synthesis of allenones via nickel-catalyzed reductive coupling of propargyl esters with acyl chlorides

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ABSTRACT

Transition metal-catalyzed reductive coupling of two electrophilic components has proved to be a powerful and step-economic tool for forging diverse organic molecules. However, the application of such strategy for the synthesis of allenones that are very important in pharmaceutical chemistry and synthetic chemistry remains very limited and challenging. Herein, we have developed a nickel-catalyzed reductive cross-coupling of propargyl esters with acyl chlorides, which affords an alternative route for the synthesis of allenones. The current catalytic protocol features readily available starting materials, wide substrate scope, mild conditions and good regioselectivity.

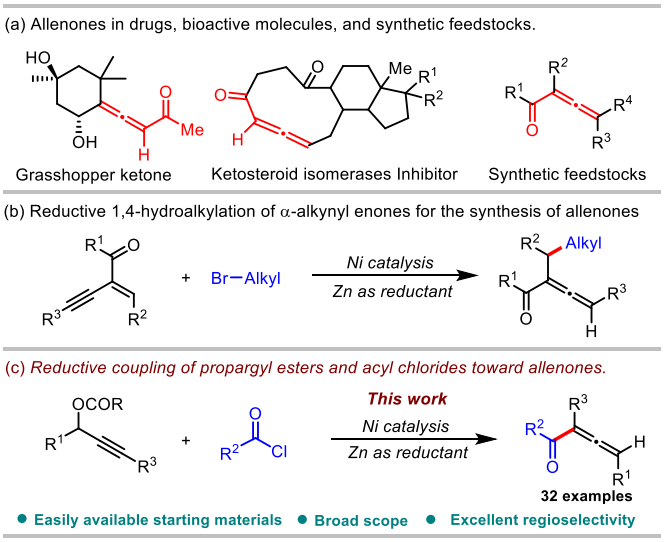
Allenones as a class of prominent structural motifs popularly exist in numerous natural products, bioactive compounds, and pharmaceuticals, and they also serve as versatile feedstocks for construction of diverse organic molecules, especially for poly-substituted heterocycles (Scheme 1a) [1,2]. Therefore, the development of methods for the efficient synthesis of functionalized allenones is of great interest and significance. The conventional routes to access allenones mainly depend on the Wittig reaction involving α-carbonyl phosphorus ylide [3,4] and the oxidation of secondary allenols or homopropargylic alcohols [5–8]. However, such methods generally suffer from tedious steps for the preparation of the starting material, the requirement of stoichiometric oxidant, and limited scope. In this regard, various catalytic approaches have been exploited as alternative strategies for the synthesis of allenones [9–11]. For instance, functionalization of 1,3-enynes with aldehydes or acyl halides, even with CO as a carbonyl source, by nickel catalysis or *N*-heterocyclic carbenes (NHCs) catalysis has recently been demonstrated as a robust platform for the synthesis of structurally diverse allenones [9f-9g, 10a-10d]. Despite these impressive advances, the facile and rapid synthesis of allenones from easily available handles remains a challenge due to the high reactivity of allenones. As such, it is highly in demand and significant to exploit new catalytic strategies for the efficient synthesis of allenones.

Transition metal-catalyzed reductive cross-coupling has emerged as one of the most powerful tools for the synthesis of organic molecules due to avoiding the use of organometallic reagents and good functional group tolerance [12]. However, despite considerable advances in this area, successful examples for the synthesis of allenones through such strategy are still very rare. Recently, Fang's group reported nickel-catalyzed reductive hydroalkylation of α-alkynyl enones with alkyl bromides to access allenones (Scheme 1b) [13a]. Wang et al. disclosed a chromium-catalyzed electrophilic coupling of propargyl bromides with aromatic aldehydes for the synthesis of allenones through sequential radical–polar crossover and Oppenauer oxidation [13b]. In consideration of easy availability and wide applications of propargyl esters in synthesis of allenes [14], as well as our continuous interest in utilization of carboxylic acid derivatives in reductive coupling [15], herein, we report the first nickel-catalyzed reductive cross-coupling of propargyl esters with acyl chlorides, affording a new alternative route for the synthesis of allenones with wide substrate scope, good functional group compatibility and regioselectivity (Scheme 1c).

Initially, Boc-protected propargyl ester **1** and aroyl chloride **2** were selected as the model substrates to optimize the reaction conditions under nickel catalysis (Table 1; also see Table S1 in supplementary information). After carefully evaluating various parameters, we found that the desired allenone **3** could be obtained in 63 % isolated yield by using $NiBr_2$·diglyme (10 mol%) as catalyst, **L1** (12 mol%) as ligand, Zn (2.0

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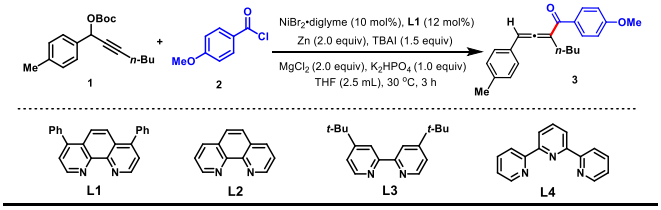


Scheme 1. Synthesis of allenones by reductive nickel catalysis.

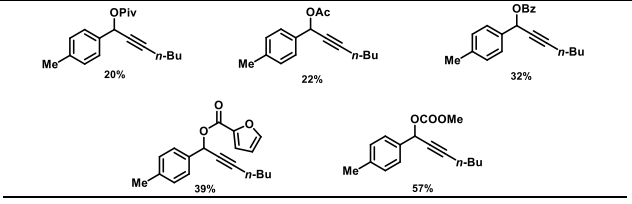
equiv.) as reductant, TBAI (tetrabutylammonium iodide), $MgCl_2$, and K_2HPO_4 as additives in THF at r.t. for 3 h (entry 1, Table 1). Subsequently, control experiments indicated that the nickel catalyst, ligand and Zn are essential for this catalytic reaction (entry 2, Table 1). Moreover, we found that no desired product was detected without the additive TBAI (entry 3, Table 1) probably due to anion exchange [16]. Besides, the $MgCl_2$ as the commonly used additive in reductive coupling [16,17] also played crucial roles in this system although its exact functions are still indefinite at this stage (entry 4, Table 1). The yield was decreased to 37 % without weak base K_2HPO_4 (entry 5, Table 1) [14d]. The other nickel catalysts such as $NiBr_2$, NiI_2 and $Ni(cod)_2$ resulted in inferior yields (entries 6–8, Table 1). Then, the performance of various ligands was examined, and it was found that the yields with substituent-free phenanthroline **L2**, bipyridine **L3** or terpyridine **L4** were lower than the case of **L1** (entries 9–11, Table 1). When Mn powder (2.0 equiv) was used as the reductant instead of Zn, the yield was decreased to 25 % (entry 12). The preliminary screening of solvent showed that almost no desired coupling product was detected in DMA, 1,4-dioxane and toluene (entry 13–15, Table 1). Additionally, we found the yield was down to 31 % with some unconfirmed side-products when the catalytic reaction was performed at 50 °C (entry 16, Table 1). It needs to be pointed out that the current catalytic system is pretty sensitive to the reaction time, and we surprisingly found that the trace amount of product **3** was observed with consumption of the product in current conditions by prolonging the reaction time to 6 h (entry 17, Table 1). This phenomenon may be explained that the allenone is indeed a highly reactive synthetic intermediate that easily undergoes various transformation under the presence of transition metal catalyst. It exactly demonstrates the potential challenges of such coupling system. Furthermore, we found that representative propargyl esters containing other protecting groups could undergo this coupling reaction, giving the coupling product **3** with lower yields (Table 1, bottom).

With the established conditions in hand, the generality of this nickel-catalyzed reductive cross-coupling of propargyl esters and acyl chlorides for the synthesis of allenones was investigated (Scheme 2). First, the scope of propargyl esters was examined. It was found that the coupling reactions of acyl chloride **2** with a series of aryl propargyl esters proceeded smoothly, affording the corresponding allenones **3**–**11** with tolerance of alkyl (**3**, **5**, **10**), ether (**6**), halides (**7**, **8**), ester (**9**), and phenyl (**11**) groups. The structure of product **3** was further confirmed by an X-ray analysis. Moreover, heterocycle-containing propargyl esters such as thiophene and carbazole were also compatible with such catalytic system, leading to the desired products **12** and **13**. Additionally, we

Table 1
Optimization of reaction conditions[a].



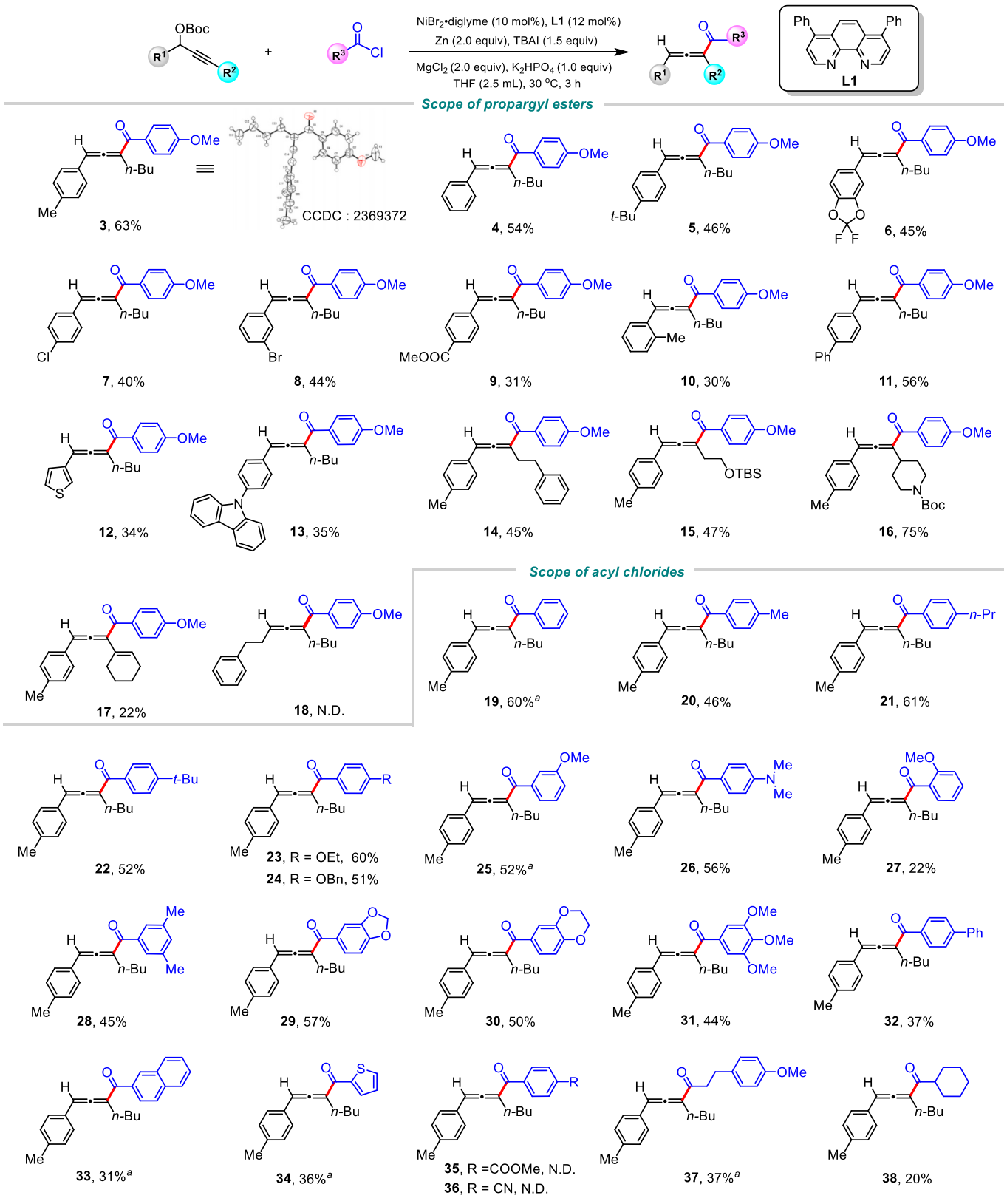
Entry	Variations from above conditions	Yield (%)
1	None	63
2	Without $NiBr_2$·diglyme/**L1**/Zn	N.D.
3	Without TBAI	N.D.
4	Without $MgCl_2$	N.D.
5	Without K_2HPO_4	37
6	$NiBr_2$ instead of $NiBr_2$·diglyme	10
7	NiI_2 instead of $NiBr_2$·diglyme	48
8	$Ni(cod)_2$ instead of $NiBr_2$·diglyme	46
9	**L2** instead of **L1**	28
10	**L3** instead of **L1**	55
11	**L4** instead of **L1**	32
12	Mn instead of Zn	25
13	DMA instead of THF	trace
14	1,4-Dioxane instead of THF	N.D.
15	Toluene instead of THF	N.D.
16	50 °C instead of r.t.	31
17	6 h instead of 3 h	trace



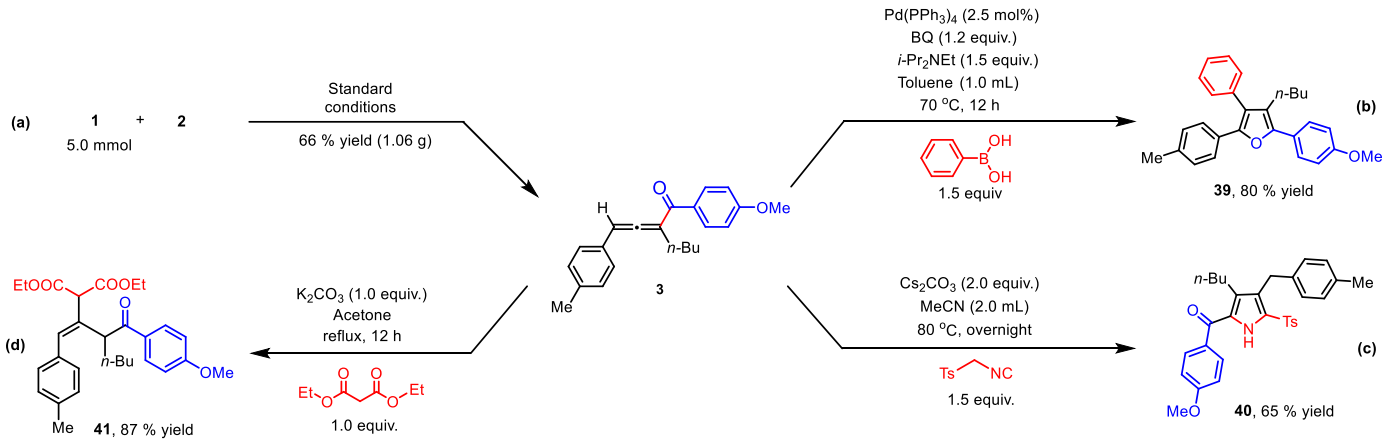
[a]Reaction conditions: **1** (0.2 mmol), **2** (2.2 equiv), $NiBr_2$·diglyme (10 mol%), **L1** (12 mol%), Zn (2.0 equiv), TBAI (1.5 equiv), $MgCl_2$ (2.0 equiv), K_2HPO_4 (1.0 equiv), and THF (2.5 mL) under N_2 at 30 °C for 3 h. Isolated yields.

found that some propargyl esters bearing diverse alkyl (**14**–**16**) and alkenyl (**17**) at terminal position of alkynyl groups smoothly reacted with acyl chloride **2**, delivering the target allenones **14**–**17**. Unfortunately, the propargyl ester with alkyl chain at propargyl position showed no reactivity, and the corresponding product **18** was not detected.

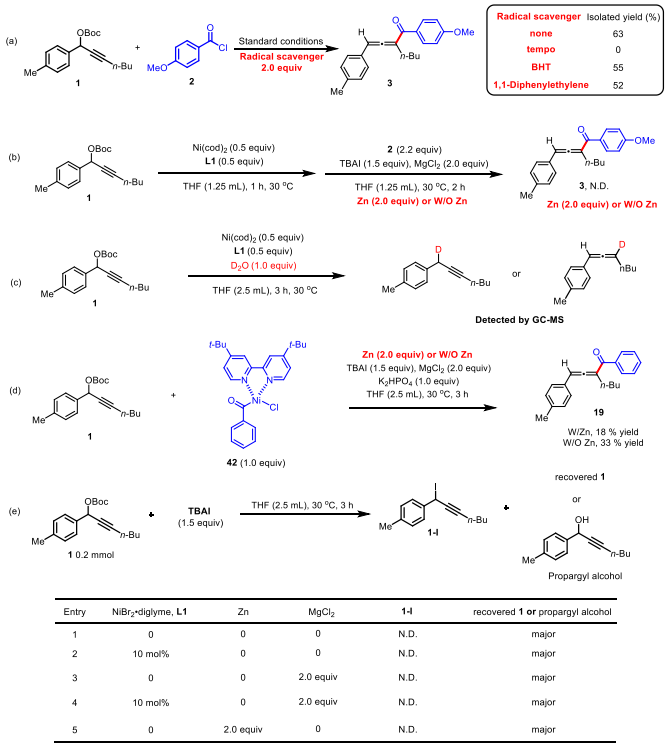
Then, the scope of acyl chlorides was evaluated. Benzoyl chloride and an array of aroyl chlorides bearing different substituents such as alkyl, ether and amino groups can smoothly undergo such coupling reactions, furnishing the corresponding allenones **19**–**26** in moderate yields. Aroyl chloride with methoxy group at *ortho* position of phenyl ring reacted with propargyl ester **1** under standard conditions, giving the desired product **27** albeit with lower yield probably due to hindrance effect. Representative poly-substituted aroyl chlorides were also amenable to this catalytic system, giving the allenones **28**–**31** in synthetically useful yields. π-Extended aroyl chlorides (**32**, **33**) and heterocyclic acyl chloride such as thiophene (**34**) could take part in this catalytic reaction, affording the corresponding products **32**–**34**. It is a pity that aroyl chlorides bearing electron-withdrawing groups such as ester and cyano were not tolerated with such protocol (**35** and **36**). Furthermore, we found that aliphatic acyl chlorides can participate in



Scheme 2. The scope of propargyl esters and acyl chlorides for this nickel-catalyzed reductive cross-coupling.
Reaction conditions: propargyl esters (0.2 mmol), acyl chlorides (2.2 equiv), $NiBr_2$·diglyme (10 mol%), **L1**(12 mol%), Zn (2.0 equiv), TBAI (1.5 equiv), $MgCl_2$ (2.0 equiv), K_2HPO_4 (1.0 equiv), and THF (2.5 mL) under N_2 at 30 °C for 3 h. Isolated Yields. [a]0 °C and 12 h.



Scheme 3. Gram-scale synthesis and synthetic applications.



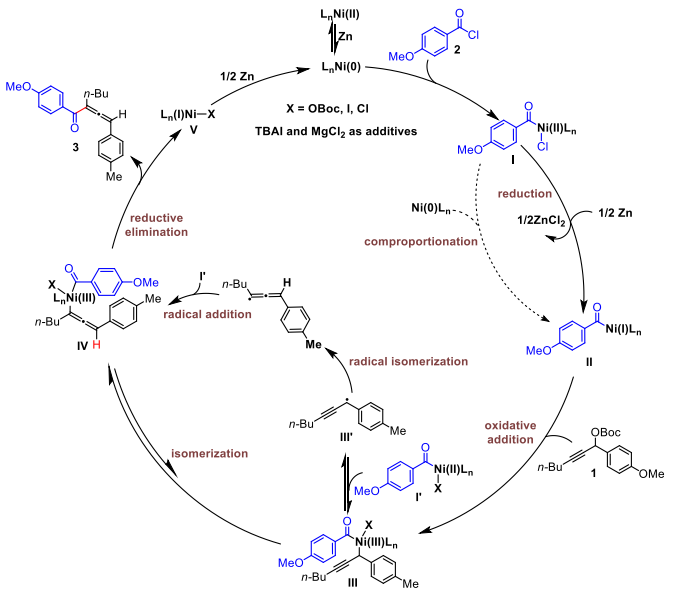
Entry	$NiBr_2$•diglyme, **L1**	Zn	$MgCl_2$	1-I	recovered **1** or propargyl alcohol
1	0	0	0	N.D.	major
2	10 mol%	0	0	N.D.	major
3	0	0	2.0 equiv	N.D.	major
4	10 mol%	0	2.0 equiv	N.D.	major
5	0	2.0 equiv	0	N.D.	major

Scheme 4. Mechanistic studies.

the reactions, providing the desired allenones such as **37** and **38**. It needs to be pointed out that lower yields for some products generally account for homo-coupling of propargyl esters and acyl chlorides as well as considerable deprotection of propargyl esters.

Furthermore, considering allenones as versatile synthons in synthetic chemistry, scale-up preparation and several synthetic applications were performed to demonstrate the potential of the current catalytic system (Scheme 3). We found that the reaction of propargyl ester **1** and acyl chloride **2** at 5.0 mmol scale under standard conditions provided the desired allenone **3** in 66 % yield (1.06 g) (Scheme 3a). Additionally, the oxidative cross-coupling of **3** with aryl boronic acid by palladium catalysis led to the corresponding poly-substituted furan **39** in 80 % yield (Scheme 3b) [18] and the metal-free cyclization reaction of **3** with isonitrile delivered the tetrasubstituted pyrrole **40** in 65 % yield (Scheme 3c) [19]. The base-mediated Michael-addition of diethyl malonate into allenone **3** afforded the β,γ-unsaturated ketone **41** in 87 % yield (Scheme 3d) [20].

Subsequently, a series of control experiments were carried out to obtain some insights into the mechanism of the current nickel-catalyzed cross-coupling. First, several radical scavengers were added to the standard conditions to examine the possibility of the radical pathway (Scheme 4a). It was found that the reaction of propargyl ester **1** with aroyl chloride **2** in the presence of 2,2,6,6-tetramethylpiperidine oxide (TEMPO) did not produce the desired coupling product **3**. However, given that TEMPO itself is a strong oxidant, this result can't confirm that the potential radical intermediates are involved in the catalytic process absolutely. Furthermore, we found that the allenone **3** could be obtained in 55 % and 52 % yields, respectively, with the addition of butylated hydroxytoluene (BHT) and 1,1-diphenylethylene, probably demonstrating non-radical pathway for the current coupling reaction. However, despite these observations, if the propargylic radical interacted with Ni intermediate faster than the radical addition, the radical trapping product would not be obtained. Therefore, the formation of radical intermediates could not be ruled out absolutely. Additionally, stoichiometric reaction of propargyl ester **1** with $Ni(cod)_2$ (0.5 equiv.) and **L1** (0.5 equiv.) was conducted in THF at 30 °C for 1 h, followed by the addition of acyl chloride **2**, TBAI (1.5 equiv.) and $MgCl_2$ (2.0 equiv.) into the reaction system with or without Zn reductant for additional 2 h, and we found that no desired product **3** was observed in both cases (Scheme 4b). In order to further investigate whether oxidative addition of propargyl ester with Ni(0) occurs, the reaction of **1** with $Ni(cod)_2$ (0.5 equiv.) was conducted in the presence of D_2O, and we found that the corresponding D-trapping alkyne or allene could be detected by GC-MS together with some homo-coupling and decomposition of starting material (Scheme 4c). This result supports the occurrence of oxidative addition to some extent, and probably homo-coupling or decomposition of propargyl ester is very fast in the presence of Ni(0), resulting in no desired cross-coupling product after the addition of acyl chloride (Scheme 4b). Subsequently, we found that the stoichiometric reaction of propargyl ester **1** with acyl-Ni (II) complex **42** prepared by oxidative addition of acyl chloride and Ni(0) and **L3** ligand could provide the corresponding product **19** albeit with relatively low yields in the presence of Zn as reductant or without it (Scheme 4d). It was therefore suggested that the oxidative addition of acyl chloride with Ni(0) to form acyl-Ni (II) species may be the initial step in the catalytic cycle. On the other hand, given that the reductant is not necessary in such reaction (Scheme 4d), it is speculated that one way to generate the Ni(I) intermediate could be achieved by the reduction of Zn, while the other way is through the comproportionation between potential Ni(0) species formed by the decarbonylation and reductive elimination of acyl-Ni (II) species in the system and Ni (II) species [21,22]. Then, the sequential oxidative addition of Ni(I) complex with the propargyl ester and reductive elimination could produce the final product. Finally, several reactions of propargyl ester **1** with TBAI under different conditions were performed,



Scheme 5. Proposed mechanism.

whereas no corresponding propargyl iodide was detected with some recover or decomposition of propargyl ester (Scheme 4e). This result probably excludes the possibility of formation of propargyl iodide in situ. In consideration of necessity of TBAI in this sytem, we proposed the potential anion exchange in the catalytic cycle should be possible.

Based on the experimental observations above and previous reports [14d,14k,14n], a plausible catalytic cycle for this nickel-catalyzed reductive cross-coupling is proposed in Scheme 5 using propargyl ester **1** and acyl chloride **2** as models. First, the oxidative addition of acyl chloride **2** with Ni(0) complex derived from Ni(II) precursor by the reduction with Zn forms an acyl-Ni(II) complex **I**. After that, two potential pathways account for the transformation of acyl-Ni(II) species **I** into the corresponding acyl-Ni(I) species **II**, namely, the reduction by Zn powder or comproportionation of acyl-Ni(II) species **I** with Ni(0). Subsequently, the oxidative addition of acyl-Ni(I) species **II** with propargyl ester **1** gives the intermediate **III**. Then, isomerization of **III** may generate the allenyl-Ni(III) species **IV**, and such regioselectivity for the formation of allenyl-Ni species is generally sterically favored [14d,14n]. Besides, it is also possible that the homolysis of intermediate **III** gives the propargyl radical **III'** and acyl-Ni(II) species **I'**, and the isomerization to allenyl radical and coupling with acyl-Ni(II) species again could generate the intermediate **IV** [23,24]. Then, the reductive elimination of species **IV** delivers the desired coupling product **3** and a Ni(I) species **V**. Finally, such Ni(I) species **V** is reduced by Zn to reproduce the Ni(0) species to complete the catalytic cycle.

In summary, the nickel-catalyzed reductive cross-coupling of propargyl esters and acyl chlorides has been achieved for the first time, which provides an alternative route for the synthesis of a variety of structurally diverse allenones. The current catalytic protocol features broad substrate scope, readily available starting materials, mild conditions and good regioselectivity. The applications are witnessed by scalability of catalytic reaction and the synthesis of representative heterocycles such as multi-substituted furan and pyrrole. Further studies on the reaction mechanism and the development of the corresponding asymmetric synthesis of such allenones will be performed in future work.

CRediT authorship contribution statement

Zhen Song: Methodology, Investigation, Formal analysis. **Yueming Wang:** Methodology, Investigation, Formal analysis. **Ming Ma:** Writing – review & editing, Validation, Supervision, Resources, Project administration. **Yuanhong Ma:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tchem.2024.100113.

Supplementary Material

Supplementary material that may be helpful in the review process should be prepared and provided as a separate electronic file. That file can then be transformed into PDF format and submitted along with the manuscript and graphic files to the appropriate editorial office.

Data availability

Data will be made available on request.

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